

A logarithmic convolution

almost every index has $\log k \approx \log n$

$$\lim_{n \rightarrow \infty} \frac{\log^2 n}{n} \sum_{k=2}^{n-2} \frac{1}{\log k \log(n-k)} = 1.$$

Overview. Put

$$x_k = \frac{\log n}{\log k}, \quad 2 \leq k \leq n-2, \quad \text{so that} \quad A_n := \frac{\log^2 n}{n} \sum_{k=2}^{n-2} \frac{1}{\log k \log(n-k)} = \frac{1}{n} \sum_{k=2}^{n-2} x_k x_{n-k}.$$

Every $x_k \geq 1$, which gives the lower bound at once. For the upper bound, Cauchy–Schwarz plus the reflection $k \mapsto n-k$ replaces the product $x_k x_{n-k}$ by x_k^2 , and then the key structural fact takes over: for any $\alpha < 1$, all but n^α of the indices satisfy $k \geq n^\alpha$, hence $\log k \geq \alpha \log n$ and $x_k \leq \frac{1}{\alpha}$. The exceptional indices are so few, a power of n below n , that even the worst possible value $x_2 = \frac{\log n}{\log 2}$ cannot rescue them.

In words: the logarithm is so slowly varying that *almost every* $k \leq n$ has $\log k$ essentially equal to $\log n$, so the sum has n terms each essentially equal to $\frac{1}{\log^2 n}$.

1. Notation and the lower bound

For $2 \leq k \leq n-2$ both $k \leq n$ and $n-k \leq n$, so $\log k \leq \log n$ and $\log(n-k) \leq \log n$, whence

$$x_k = \frac{\log n}{\log k} \geq 1, \quad x_{n-k} = \frac{\log n}{\log(n-k)} \geq 1. \quad (1)$$

Therefore each of the $n-3$ summands of

$$A_n = \frac{1}{n} \sum_{k=2}^{n-2} x_k x_{n-k} \quad (2)$$

is at least 1, and

$$A_n \geq \frac{n-3}{n}. \quad (3)$$

In particular $\liminf_n A_n \geq 1$. The whole problem is the matching upper bound.

2. Removing the convolution by Cauchy–Schwarz

Apply the Cauchy–Schwarz inequality to (2):

$$\sum_{k=2}^{n-2} x_k x_{n-k} \leq \sqrt{\left(\sum_{k=2}^{n-2} x_k^2 \right) \left(\sum_{k=2}^{n-2} x_{n-k}^2 \right)}.$$

The substitution $k \mapsto n-k$ is a bijection of the index range $\{2, \dots, n-2\}$ onto itself, so the two factors are equal and

$$A_n \leq B_n := \frac{1}{n} \sum_{k=2}^{n-2} x_k^2 = \frac{\log^2 n}{n} \sum_{k=2}^{n-2} \frac{1}{\log^2 k}. \quad (4)$$

By the same reasoning as (1), also $B_n \geq \frac{n-3}{n}$. The convolution structure has been removed at no cost: it remains to show $B_n \rightarrow 1$.

3. The upper bound: splitting at n^α

Fix $\alpha \in (0, 1)$ and split the index range at n^α :

$$B_n = \frac{1}{n} \sum_{2 \leq k < n^\alpha} x_k^2 + \frac{1}{n} \sum_{n^\alpha \leq k \leq n-2} x_k^2. \quad (5)$$

Small indices. There are fewer than n^α of them, and the largest possible value of x_k on the whole range is $x_2 = \frac{\log n}{\log 2}$. Hence

$$\frac{1}{n} \sum_{2 \leq k < n^\alpha} x_k^2 \leq \frac{n^\alpha}{n} \left(\frac{\log n}{\log 2} \right)^2 = n^{\alpha-1} \left(\frac{\log n}{\log 2} \right)^2 \xrightarrow{n \rightarrow \infty} 0, \quad (6)$$

since $\alpha - 1 < 0$ and a power beats a square of a logarithm.

Large indices. If $k \geq n^\alpha$ then $\log k \geq \alpha \log n$, so $x_k \leq \frac{1}{\alpha}$ and

$$\frac{1}{n} \sum_{n^\alpha \leq k \leq n-2} x_k^2 \leq \frac{1}{n} \cdot n \cdot \frac{1}{\alpha^2} = \frac{1}{\alpha^2}. \quad (7)$$

Now let $\varepsilon > 0$. Choose $\alpha < 1$ close enough to 1 that

$$\frac{1}{\alpha^2} < 1 + \frac{\varepsilon}{2},$$

and then N so large that for $n \geq N$ both

$$n^{\alpha-1} \left(\frac{\log n}{\log 2} \right)^2 < \frac{\varepsilon}{2} \quad \text{and} \quad \frac{n-3}{n} > 1 - \varepsilon.$$

Combining (3), (4), (5), (6), (7), for every $n \geq N$

$$1 - \varepsilon < \frac{n-3}{n} \leq A_n \leq B_n \leq \frac{\varepsilon}{2} + \frac{1}{\alpha^2} < 1 + \varepsilon.$$

Since ε was arbitrary,

$$\boxed{\lim_{n \rightarrow \infty} B_n = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} A_n = 1.}$$

Note the order of the choices: α is selected from ε first, and only then is N selected from both. The exponent α is a free parameter that trades the two error terms against each other: pushing α towards 1 improves (7) and worsens (6), but (6) has a power of n to spend and always wins.

4. How slow the convergence is

The proof gives no useful rate, and for good reason: the true rate is logarithmic.

n	A_n	$A_n - 1$	$2/\log n$
10^2	1.7792	0.7792	0.4343
10^3	1.4603	0.4603	0.2895
10^4	1.3002	0.3002	0.2171
10^5	1.2204	0.2204	0.1737

At $n = 10^5$ the sequence is still 22% above its limit. The reason is visible from a continuous model. Write $k = tn$ with $t \in (0, 1)$ and $L = \log n$; then

$$\log k = L + \log t, \quad x_k = \frac{L}{L + \log t} = \frac{1}{1 + \frac{\log t}{L}} \approx 1 - \frac{\log t}{L},$$

and similarly for x_{n-k} with t replaced by $1 - t$. Hence

$$x_k x_{n-k} \approx 1 - \frac{\log t + \log(1 - t)}{L},$$

and averaging over $t \in (0, 1)$, using $\int_0^1 (-\log t) dt = \int_0^1 (-\log(1 - t)) dt = 1$,

$$\boxed{A_n \approx 1 + \frac{2}{\log n}.} \quad (8)$$

The last column of the table is this prediction. It captures the right order of magnitude and the right constant asymptotically, though at these values of n the neglected $O(\log^{-2} n)$ term is still comparable to the main correction, which is what one expects when the expansion parameter is $1/\log n \approx 0.09$ at $n = 10^5$.

5. What made it work

The only property of $\log$ used in Section 3 was

$$\log(n^\alpha) = \alpha \log n, \quad (9)$$

i.e. that the logarithm of a fixed power of n is a fixed fraction of $\log n$. This is what forces x_k to be uniformly bounded on all but a negligible set of indices, and it is a statement about slow variation, not about the logarithm specifically.

Weight $w(k)$	Behaviour of $\frac{w(n)^2}{n} \sum_k \frac{1}{w(k)w(n-k)}$
$w(k) = \log k$	Limit 1, by the argument above.
$w(k) = \log^p k$, $p > 0$	Limit 1: $\left(\frac{\log n}{\log k}\right)^p \leq \alpha^{-p}$ on $k \geq n^\alpha$, and the small-index block is still killed by $n^{\alpha-1} \log^{2p} n \rightarrow 0$.
$w(k) = k^\beta$, $\beta > 0$	Not 1: here $\frac{w(n)^2}{w(k)w(n-k)} = (t(1-t))^{-\beta}$ does not concentrate, and the limit is the Beta integral $\int_0^1 t^{-\beta}(1-t)^{-\beta} dt = B(1-\beta, 1-\beta)$ for $0 < \beta < 1$, diverging for $\beta \geq 1$.

The contrast in the last row is the point of the problem. For a power weight the rescaled summand is a genuine function of $t = k/n$ and the limit is a Riemann integral, generally not 1. For a logarithmic weight the rescaled summand converges to the constant 1 for *every* fixed $t \in (0, 1)$, and the limit is $\int_0^1 1 dt = 1$. Formula (8) is the first correction to that degeneracy, and it is the Riemann integral of $-\log t - \log(1 - t)$, divided by $\log n$.